Experiment – 1 : Comparative Study of Low-Code Development Versus Traditional Application Development Approaches

Aim:

To compare low-code development and traditional application development approaches based on development speed, ease of use, customization, cost, maintenance, scalability, and overall suitability for different types of software projects.

Theory:

Application development is the process of designing, developing, testing, and deploying software applications. There are two major approaches to application development:

Low-Code Development:
Low-code development platforms provide graphical user interfaces, drag-and-drop components, and pre-built templates that allow developers to build applications with minimal manual coding. These platforms reduce development time and enable both professional developers and non-technical users (citizen developers) to create applications quickly. Low-code is commonly used for business process automation, internal tools, and rapid application development.

Traditional Application Development:
Traditional development involves writing source code using programming languages such as Java, Python, C#, or JavaScript. Developers have complete control over application design, architecture, and functionality. Although this approach requires more programming knowledge and development time, it offers greater flexibility, scalability, security, and customization for complex software systems.

Procedure:

1. Study the concepts of low-code development and traditional application development.
2. Select a low-code platform and review its features, including drag-and-drop interface, templates, and automation tools.
3. Study the traditional software development process using programming languages and frameworks.
4. Compare both approaches based on development time, coding effort, customization, scalability, cost, maintenance, and deployment.
5. Record the observations in a comparison table.
6. Analyze the advantages and limitations of each approach.
7. Draw conclusions regarding the most suitable development approach for different application requirements.

Result:

The comparative study shows that low-code development significantly reduces development time and coding effort, making it suitable for rapid application development and business applications. Traditional application development, while requiring greater time and programming expertise, provides superior flexibility, scalability, performance, and customization, making it the preferred choice for large-scale and complex software systems.